Technology behind evolving robotic hands and Japanese companies

TECHNOLOGY / ROBOTICS | Approaching the human hand

A robotic hand needs not only drives to move its fingers, but precision parts supporting joints, sensors measuring force and tactile technology detecting contact and slipping. Combined with control technology, these allow it to handle fragile objects and objects of different shapes.

From gripping to handling with controlled force

To broaden tasks in logistics, shops and homes, hands must adapt to shape, hardness and weight, beyond gripping predefined objects. In addition to finger count, important factors include force control, slip detection, compactness, light weight, power consumption, durability and mass-production cost.

Six elements behind hand motion

Skeleton/joints: bearings and precision mechanical parts support smooth motion.
Drives: small motors and actuators move the fingers.
Power transmission: reducers and similar parts adjust speed and force.
Force sensing: force/torque sensors measure applied force.
Touch: detects pressure and slipping to identify contact conditions.
Control: AI and other systems adjust movement based on sensor data.

Related Japanese companies and business challenges

TECHNOLOGY / ROBOTICS

MinebeaMitsumi | 6479

Makes miniature bearings, actuators, strain gauges and other products. Works with Harmonic Drive Systems on joint development of robotic hands.

Harmonic Drive Systems | 6324

Offers precision reducers and actuators. Has technology related to miniaturizing and improving the accuracy of joints, including robotic-hand drives.

THK | 6481

In addition to linear-motion parts, offers picking robot hands and suction hands. Works in end-effector technology for handling goods in logistics and production.

NSK | 6471

Makes bearings, precision mechanical parts, actuators and other products. Compactness, light weight and power transmission are important technologies for robotic joints and end effectors.

NISSHA | 7915

Offers thin pressure-sensing technology and related products, a relevant field for giving robotic fingers and palms the ability to sense contact.

Separate market growth from company results

Even if the market grows, business effects vary by adopting customers, mass-production timing, unit prices and margins. Prototype development and volume adoption are different stages. At companies with large non-robotics businesses, adoption need not immediately drive rapid company-wide growth.